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Cross-Instrument Source Detection and Astrometry with a Joint Rubin–Euclid Foundation Model

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ABSTRACT

The Vera C. Rubin Observatory and *Euclid* survey overlapping sky at complementary wavelengths and resolutions, but their pixel-level data are rarely combined before catalog construction. We present Jaisp, a joint Rubin–*Euclid* foundation model that ingests six Rubin optical bands (*ugrizy*) together with *Euclid* VIS and NISP (*YJH*) imagery at a common fused resolution, and learns a shared representation by masked reconstruction. We use the frozen representation to drive two downstream tasks on ∼790 paired tiles in the ECDFS field: source detection with a center-prediction head, and astrometry via a multi-band joint centroid that fuses the sharp *Euclid* VIS localization across all instruments. The joint multi-band position reduces per-band centroid scatter relative to a single-band measurement, with the largest gains at low signal-to-noise (∼2.4–2.5× over a VIS-only anchor at SNR = 5 in a controlled source-injection test), and is validated against *Gaia* to 4.4–4.8 mas at $G < 18.5$. Aggregating the per-band residuals across the field, we recover a coherent degree-scale ∼4–7 mas concordance field between the Rubin and *Euclid* astrometric solutions that is stable across foundation-model versions and reconstruction settings — a direct map of the residual cross-survey WCS mismatch that classical per-instrument pipelines leave unmodeled. Through ablations and injection tests we find the cross-instrument gain is genuine but bounded by the VIS localization floor, and we characterize a completeness loss near bright stars traceable to the network's signal-to-noise input normalization. Finally we report a first out-of-distribution test, applying the frozen model without retraining to the EDF-S field: detection and *Euclid* NISP astrometry transfer cleanly, [TODO: state final Rubin optical transfer result after real-variance rerun]. [TODO: one-sentence takeaway on the value of multi-instrument pixel-level fusion.]

Keywords: Astrometry — Surveys — Convolutional neural networks — Sky surveys — Astronomy data analysis

1. INTRODUCTION

Rubin and *Euclid* image overlapping sky with complementary strengths: Rubin provides deep six-band optical photometry at ground-based resolution, while *Euclid* VIS delivers diffraction-limited optical morphology and NISP adds near-infrared coverage. Jointly they constrain spectral energy distributions and positions better than either alone, yet standard pipelines process each instrument independently and combine only at the catalog level, discarding pixel-level cross-instrument information.

[TODO: Position relative to prior astronomical foundation / multimodal representation models (e.g. AstroCLIP, Multimodal Universe, AstroPT) and to classical multi-instrument forced photometry / forced astrometry. Make the novelty boundary explicit: pixel-level, multi-instrument, mixed-resolution fusion driving downstream detection and astrometry.]

In this paper we (i) describe Jaisp, a joint Rubin–*Euclid* foundation model trained by masked reconstruction on a common fused pixel grid (§3); (ii) use the frozen representation for source detection and for a multi-band joint-centroid astrometry that transfers the sharp *Euclid* VIS localization to bands where the source is faint (§4); (iii) validate positions against truth via source injection and against *Gaia*; and (iv) report a first out-of-distribution evaluation on a second field with no retraining (§5). Throughout we take a deliberately critical stance, using ablations and controlled injections to separate what the multi-instrument representation genuinely buys from what is already available in the single sharpest band.

Our main contributions are:

1. A multi-instrument, mixed-resolution pixel-level foundation model for Rubin+*Euclid*, with frozen-feature downstream heads.

2. A multi-band joint-centroid astrometry method and a controlled quantification, via source injection, of the cross-instrument localization gain and its VIS-imposed ceiling.

3. A measurement of a coherent degree-scale concordance field between the Rubin and *Euclid* astrometric solutions, shown to be stable across foundation-model versions and fitting choices.

4. A first transfer (out-of-distribution) evaluation of the frozen stack on an independent field.

5. Honest characterization of failure modes: foundation-version insensitivity, the VIS localization floor, and bright-star detection suppression.

2. DATA

Our primary field is ECDFS (Rubin tract 5063), where we assemble ∼790 paired Rubin+*Euclid* tiles. [TODO: exact tile count, sky area, depth, epoch.] For the out-of-distribution test (§5) we use 72 paired tiles in the EDF-S field (tract 2394).

Each tile carries six Rubin bands (*ugrizy*) and four *Euclid* bands (VIS, NISP Y, J, H). *Euclid* imagery is taken from the MER mosaics; VIS and NISP are sampled at $0.1''\,\mathrm{px}^{-1}$ (∼1084×1084 per tile), and Rubin at [TODO: Rubin pixel scale / tile shape (6×512×512)]. The native pixel scales and per-band World Coordinate Systems are retained; the foundation model resamples internally to a common fused scale of $0.4''\,\mathrm{px}^{-1}$ (§3).

Per-pixel variance maps accompany the Rubin imagery and, in the primary field, the *Euclid* imagery. [TODO: Note: the EDF-S *Euclid* tiles used for the first OOD pass lacked variance maps; see §5. State final data state after real-variance tiles are obtained.]

For astrometric validation we use *Gaia* [TODO: DR] as an absolute reference, and [TODO: the *Euclid* MER catalog for detection completeness/purity]. We additionally construct a controlled source-injection set (§4) to measure positional error against known truth.

3. METHODS

3.1. *The* JAISP *foundation model*

The current design follows from a sequence of approaches we tried and discarded, and the path is itself one of the paper's lessons. We first pursued *latent-space alignment* objectives that pull co-located Rubin and *Euclid* features together: patch-level contrastive learning, an object-centric DETR-style variant with learned object queries, and JEPA-style student–teacher matching at the token level, including a strict-position variant. Each failed for precision work, for a common reason. Contrastive patch embeddings collapsed onto the sky background that dominates ∼95% of the pixels; the object-centric matching was unstable in crowded deep fields; and the token-level objectives, even with exact positional matching, rewarded feature *similarity* without requiring the encoder to preserve where flux sits to sub-pixel accuracy — the coarse (∼3″) patch tokenization compounded this. The decisive evidence was that a simple supervised cost-volume CNN reached ∼38 mas on the astrometry task while the best JEPA features gave only ∼47 mas: the expensive self-supervised representation was worse than a small supervised baseline. We concluded that latent-similarity objectives do not retain the spatial information precision cosmology needs, and moved to *pixel-space masked reconstruction*, where holding out a band and reconstructing its pixels forces the encoder to encode exact positions. A fuller account of the abandoned variants is given in the project documentation.

JAISP encodes each band with a per-band convolutional stem, resamples the per-instrument streams to a common fused pixel scale of $0.4''\,\mathrm{px}^{-1}$, and fuses them into a shared bottleneck representation. [TODO: Architecture detail: per-band stems, mixed-resolution stream depths from $\log_2$(scale ratio), ConvNeXt/transformer blocks, FiLM band conditioning, concat fusion (v9/v10).] The per-band stem normalizes its input by the per-pixel noise, $x = \mathrm{image}/\mathrm{rms}$, and clamps the result to a fixed signal-to-noise range before feature extraction; this choice is revisited in §4 as the origin of bright-star detection suppression.

The model is trained by masked reconstruction [TODO: (MAE objective, masking fraction, cross-instrument masking, Charbonnier+core-L2 loss for v10), epochs, optimizer, hardware]. All downstream heads operate on *frozen* features.

Within the reconstruction era the architecture was refined in several steps: an initial dense variant downsampled *Euclid* VIS to the Rubin grid and discarded its resolution advantage; this was fixed by a mixed-resolution two-stream encoder that keeps each instrument at native scale, followed by a finer fused scale ($0.4''\,\mathrm{px}^{-1}$ via cropped inputs), symmetric per-band concatenation fusion, and a Charbonnier reconstruction loss with extra weighting on high-signal core pixels. We report the production checkpoint used throughout (the loss-shaped, concat-fusion variant). In-distribution probes and downstream detection/astrometry metrics do not

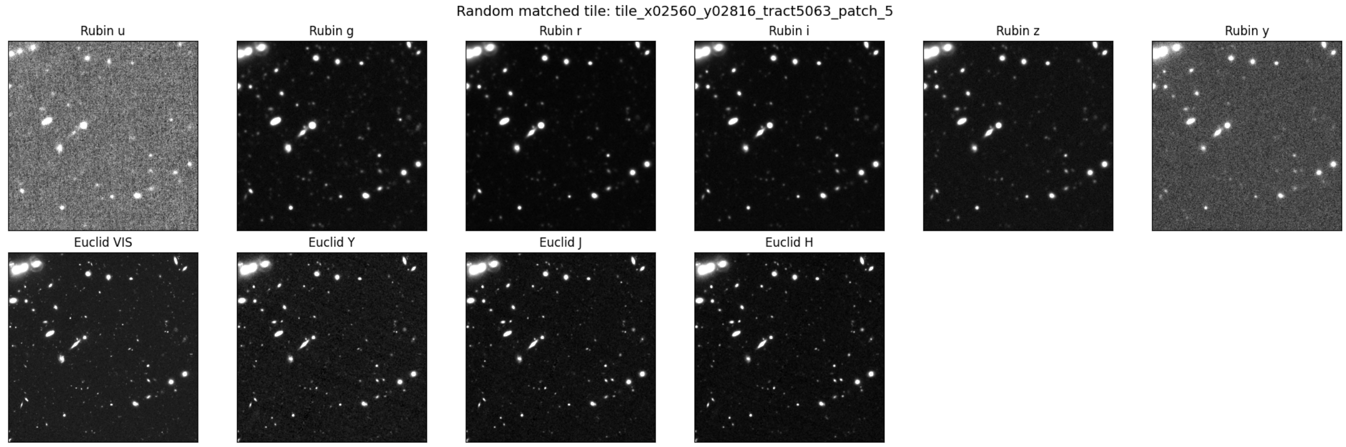

Figure 1. The ten input channels for a single matched tile. Top: Rubin *ugrizy* (512^2 at $0.2'' \, \mathrm{px}^{-1}$). Bottom: *Euclid* VIS and NISP *YJH* ($\sim 1084^2$ at $0.1'' \, \mathrm{px}^{-1}$, from the MER mosaics). The sharp VIS channel and the depth and noise differences across bands are the raw material the foundation model fuses.

statistically distinguish the last few variants (§6); we therefore do not claim a winner among them on this field and treat out-of-distribution transfer as the discriminating test. [TODO: Fix final version nomenclature/checkpoint name and training config.]

3.2. *Detection head*

We detect sources with a center-prediction (CenterNet-style) head on the frozen fused features, producing a normalized centroid and score per source. [TODO: Head architecture, self-training/pseudo-label procedure, operating point (conf threshold), NMS.] Detections are gated on the *Euclid* VIS signal and serve as seeds for astrometry.

3.3. *Multi-band joint-centroid astrometry*

Given a detection seed, we (i) refine the centroid classically on *Euclid* VIS, (ii) solve for a single *joint canonical* sky position by simultaneously fitting the source across all bands, weighting each band by its noise and accounting for its PSF, and (iii) measure each band's classical centroid and record its offset from the joint canonical position. The joint canonical position is the cross-instrument estimate; the per-band offset is the astrometric residual we report.

Because the fit is dominated by the band with the sharpest, highest-SNR localization (typically *Euclid* VIS), the joint position transfers that localization to bands in which the source is faint or unresolved (§4). Per-source uncertainties follow from the χ^2 curvature of the joint fit and a per-band FWHM/(2.355 SNR) centroid-noise model.

We emphasize that step (i)–(iii) are classical given the detection seeds; the foundation's role in astrometry is to supply the cross-instrument detections. A learned per-object position head was also trained and is discussed as an ablation in §6.

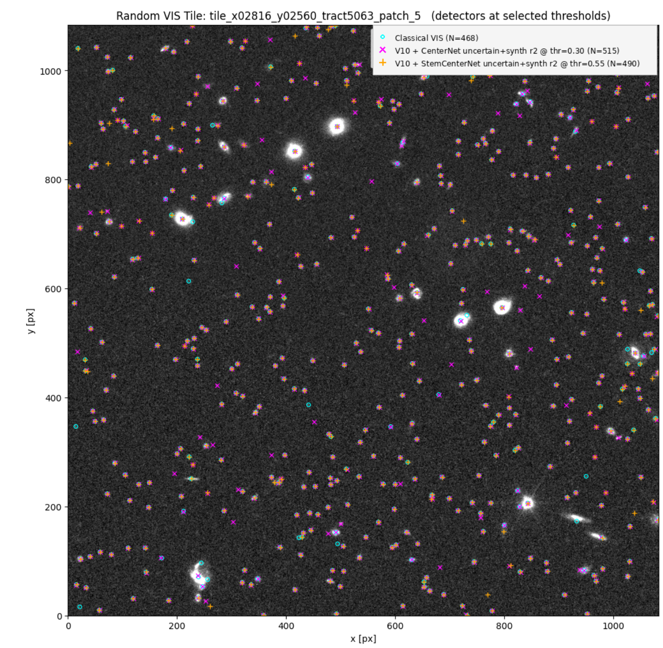

Figure 2. Detections on a representative *Euclid* VIS tile at the selected operating point: the classical VIS reference, the fused center-prediction head, and the native-resolution stem head. [TODO: Tighten legend / crop; add a zoom-in panel near a bright star to motivate §4.1.]

4. RESULTS ON ECDFS

4.1. *Detection*

On the ∼790 ECDFS tiles the detector returns ∼413 sources per tile at the chosen operating point (Fig. 2). [TODO: Completeness/purity against a reference catalog; comparison to a classical VIS baseline (notebook 06).]

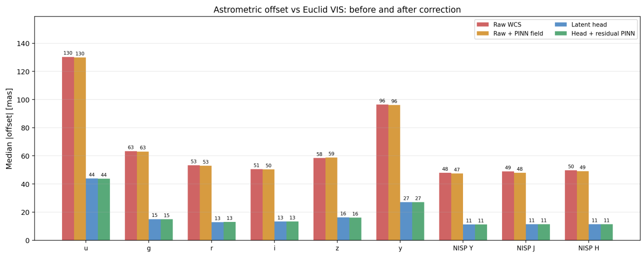

Figure 3. Median per-band astrometric offset to the *Euclid* VIS reference, before and after correction. The raw cross-instrument offset (red) ranges from ∼50 mas in the red optical bands to 130 mas in u; the per-object correction (blue) brings every band to ∼11–16 mas, with NISP reaching 11 mas. The smooth concordance field alone (orange) removes little of the per-source scatter, confirming the residual is dominated by object-level centering, not a bulk WCS term. [TODO: Confirm whether the per-band panel should show the joint-centroid instrument rather than the latent-head ablation shown here (see text).]

Detection completeness is reduced near bright stars: the source count per tile anticorrelates with the fraction of the tile under bright extended flux (corr $= -0.46$). We trace this to the per-band stem normalization (§3.1): a saturated star produces a wide region pinned at the clamped signal-to-noise ceiling, which dominates the group-wise feature normalization and depresses faint-source activations. We verify this is *not* a removable smooth background (subtracting a spatially varying background changes the count negligibly) and that it is intrinsic to the detector rather than a noise-model artifact. [TODO: Quantify against MER truth: are the lost sources real?]

4.2. *Astrometry: per-band residual and SNR scaling*

The joint-centroid astrometry reduces the pooled per-band median residual from ∼50 mas (raw band-vs-canonical, dominated by per-band centroid scatter) to ∼11–16 mas per band (∼11 mas pooled in the NISP and red optical bands; Fig. 3). The residuals follow the expected FWHM/(2.355 SNR) centroid-noise scaling (Fig. 4), reaching [TODO: floor at high SNR].

4.3. *Source-injection truth test*

To measure positional error against known truth we inject sub-pixel point sources across all ten bands at a range of signal-to-noise levels and recover their positions with the same pipeline. At SNR $= 5$ (Rubin) the classical single-band centroid errs by ∼47 mas, the joint/canonical position by ∼18–19 mas, approaching the VIS-only localization floor of ∼17 mas; a position fitted jointly across all bands improves on a VIS-only anchor by ∼2.4–2.5× (16.6 → 6.7 mas at peak SNR $= 5$).

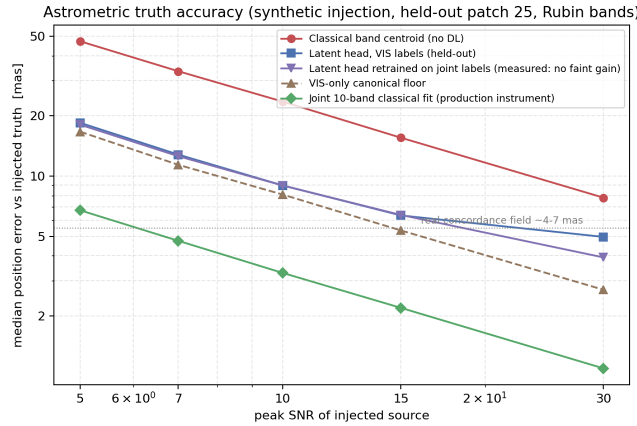

Figure 4. Astrometric error versus true position from controlled source injection into held-out tiles (Rubin bands). The classical single-band centroid (red) is the no-fusion baseline; the joint multi-band position approaches the *Euclid* VIS-only localization floor (brown dashed), and the joint ten-band classical fit (green) is the production position instrument. The latent-head curves (blue/purple) overlap at the faint end, showing the head's ∼17–18 mas floor is intrinsic, not label-limited. The dotted line marks the ∼4–7 mas scale of the real concordance field (§4.5).

[TODO: Confirm numbers, add table, state caveats: idealized Gaussian PSFs, isolated sources, injected NISP sharper than the real MER PSF.] The test also bounds the gain: where a band is intrinsically sharper than VIS, transferring the VIS position degrades rather than improves it (§6).

4.4. *Absolute validation against Gaia*

Cross-matching the joint canonical positions to *Gaia* gives a bulk offset of $(+3.7, +1.9)$ mas and a per-star scatter of ∼5.4 mas after bulk removal, improving to 4.4–4.8 mas for $G < 18.5$. [TODO: Gaia DR, matching radius, N, magnitude-binned table, spatial structure of the offset.]

4.5. *A coherent cross-survey concordance field*

After removing the bulk offset, the per-band residuals between the Rubin and *Euclid* astrometric solutions are not spatially random: they trace a smooth, coherent field across the survey footprint. Decomposing the matched source offsets with a multi-scale field solver, we isolate a degree-scale component of amplitude ∼4–7 mas that varies slowly across the field (Fig. 5). This is the residual mismatch between the two instruments' independently solved World Coordinate Systems — precisely the unmodeled structure that classical per-instrument pipelines, which calibrate each survey's astrometry separately, leave in place (§1, SITCOMTN-159).

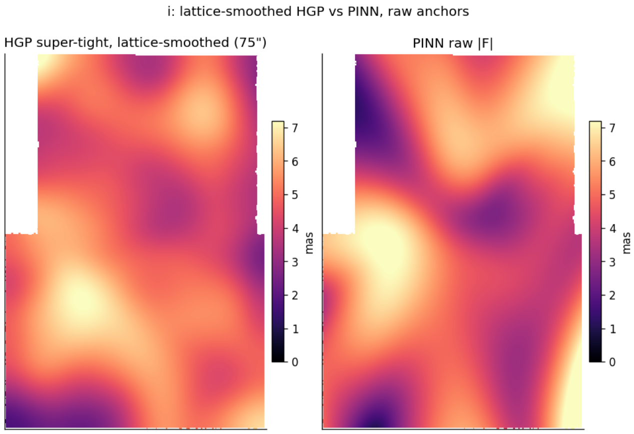

Figure 5. The degree-scale cross-survey concordance field (i band), recovered two independent ways from the matched per-band residuals: a lattice-smoothed hierarchical-GP fit (left) and a physics-informed smooth-field fit (right). Both recover the same ∼4–7 mas large-scale topology; their agreement at this scale — and across foundation-model versions — is the evidence that the field is a property of the data rather than solver noise. [TODO: Optionally add the cross-version overlay / correlation panel.]

We establish that this degree-scale field is a real measurement and not solver noise: it is recovered consistently across independent fits, across the choice of duplicate handling, across reference catalogs, and — importantly — across Jaisp foundation-model versions, with the smoothed field agreeing at Pearson r up to 0.92. [TODO: State the smoothing scale (∼150″) and the cross-comparison numbers; reference the controlled refit audit.]

We are deliberate about what is and is not established. Only the degree-scale component is reproducible; sub-degree structure in any single solver product sits below the solver's run-to-run repeatability floor (∼5 mas) and is not interpretable. We therefore report the field as a characterized *measurement* of the cross-survey WCS residual, not as a production correction applied to the catalog.

5. OUT-OF-DISTRIBUTION TRANSFER: EDF-S

The value proposition of a foundation model is transfer without retraining. All results above are in the ECDFS field on which the downstream heads were tuned. Here we apply the frozen Jaisp stack — foundation, detector, and joint-centroid astrometry — to 72 paired tiles in the independent EDF-S field (tract 2394) with no retraining or refitting.

[TODO: State the final data condition. The first pass used *Euclid* tiles without variance maps, requiring a median-absolute-deviation RMS fallback for VIS/NISP (Rubin variance was present and used directly). Results

Figure 6. Per-band and SNR-binned astrometric residual, EDF-S (OOD) versus ECDFS. [TODO: final figure from notebook 15.]

below are either (a) to be reported on real-variance tiles, or (b) reported with the fallback and the real-variance rerun as a robustness check.]

Detection transfers cleanly (456 sources per tile versus 413 on ECDFS; no processing errors), and *Euclid* NISP astrometry matches the in-distribution residual (35–41 mas versus 40 mas). The signal-to-noise scaling of the residual is reproduced. [TODO: Final Rubin optical ($g\,r\,i\,z$) transfer result: the first (fallback) pass showed a ∼1.5× larger matched-SNR residual whose origin — the RMS fallback perturbing the joint-fit weighting versus genuine field differences — is resolved by the real-variance rerun. Report the resolved outcome here.]

The bright-star detection suppression (§4.1) reproduces and is somewhat stronger in EDF-S (corr $= -0.71$), consistent with its origin in the input normalization rather than the field. [TODO: MER-catalog completeness in EDF-S.]

6. DISCUSSION

6.1. *What the multi-instrument representation actually buys*

The cross-instrument gain is real but bounded. The joint position improves faint-band localization because it imports the sharpest available (*Euclid* VIS) constraint; it adds no precision beyond that band, and where another band is intrinsically sharper, transferring the VIS position is counterproductive (§4.3). The practical value is therefore largest exactly where it is most needed: measuring positions in bands where a source is too faint to localize on its own.

6.2. *The cross-survey concordance field*

The degree-scale concordance field (§4.5) is, to our knowledge, the first direct pixel-level map of the residual astrometric mismatch between the Rubin and *Euclid* WCS solutions over a deep field. Its stability across foundation-model versions and fitting choices is what makes it credible: a structure that survives independent refits, duplicate-handling choices, and a change of backbone is a property of the data, not of any one solver. This connects directly to the motivation of §1 — the unmodeled, slowly varying residuals that single-instrument pipelines cannot see by construction, because each survey is calibrated against external references in isolation rather than against the other. We stress the field's current status as a *measurement* rather than a correction: only the degree-scale component clears the solver's re-

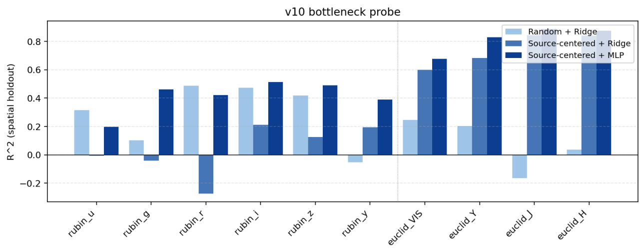

Figure 7. Per-band linear/MLP probe of the frozen bottleneck (spatial holdout). Source-centered probes recover the *Euclid* channels at $R^2 \sim 0.8$; the recovered content is statistically equivalent across foundation versions (v8/v9/v10). [TODO: Optionally replace with the side-by-side v8/v9/v10 comparison to make equivalence explicit.]

peatability floor, and applying the field as a per-source correction does not, at present, improve held-out positions (§4.5). Turning the measured field into a validated cross-survey astrometric correction — and testing whether it persists in other fields — is a natural next step.

6.3. *Foundation-version insensitivity*

We trained several foundation variants (§3.1). In-distribution linear probes and downstream detection/astrometry metrics do not statistically distinguish them. [TODO: Quantify.] The bottleneck does carry substantial extractable per-band content — source-centered probes recover the Euclid channels at $R^2 \sim 0.8$ and the optical channels well above chance (Fig. 7) — but this content is equivalent across versions. We argue the discriminating test for foundation quality is out-of-distribution transfer (§5) rather than in-distribution downstream accuracy, which saturates against label noise.

6.4. *Label noise and the limits of in-distribution metrics*

[TODO: Summarize the centroid-emulation finding: at the faint end the band-vs-canonical residual measures the model reproducing the canonical centroid definition, not positional truth; this is why injection and *Gaia* (truth metrics) are necessary and in-distribution residuals are not sufficient.]

6.5. *Bright-star detection suppression*

The signal-to-noise input normalization that stabilizes training also lets a saturated star dominate the local feature statistics, suppressing faint detections nearby (§4.1). This is a completeness, not an astrometric-accuracy, effect. Mitigations — local background subtraction, a bright-star-aware rescue pass, or a normalization less sensitive to saturated outliers — are left to future work. [TODO: If MER shows the lost sources are real, quantify the completeness cost and any mitigation tested.]

6.6. *Limitations*

- Injection uses idealized PSFs and isolated sources; blends and realistic wings are not captured.
- Out-of-distribution evidence is from a single second field.
- [TODO: variance-map availability; Rubin optical transfer; others.]

7. SUMMARY

We presented JAISP, a joint Rubin–*Euclid* foundation model operating on a common fused pixel grid, and used its frozen representation for source detection and multi-band joint-centroid astrometry. Our main findings:

1. A joint multi-band position transfers the sharp *Euclid* VIS localization to faint bands, improving low-SNR positions by ∼2.4–2.5× over a single-band measurement in controlled injection tests, and validating against *Gaia* to 4.4–4.8 mas at $G < 18.5$.
2. The gain is bounded by the VIS localization floor; the representation adds no precision beyond the sharpest contributing band.
3. The per-band residuals reveal a coherent degree-scale (∼4–7 mas) concordance field between the Rubin and *Euclid* astrometric solutions, stable across foundation versions and fitting choices — a measurement of cross-survey WCS residuals that single-instrument pipelines cannot access.
4. The frozen stack transfers to an independent field without retraining for detection and *Euclid* NISP astrometry; [TODO: final Rubin optical transfer statement.]
5. Detection completeness is suppressed near bright stars by the input normalization, a characterized and mitigable effect.

[TODO: Closing sentence on the role of pixel-level multi-instrument fusion for Rubin+*Euclid* survey science, and next steps (additional fields, photometry, weak-lensing shapes).]

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Software: Astropy (**?**), NumPy, SciPy, PyTorch, SEP (**?**)